

Weighbridge Management System

Product Manual

Version 1.0.0 · Truck weighing, gate entry, plate recognition and Oracle integration

Prepared for	
Site	
Installed by	
Date	

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1. Introduction

The **Weighbridge Management System** records every truck that is weighed on your weighbridge(s), controls the gate barriers, and passes the completed records to your Oracle database. It replaces manual registers and paper slips with a single screen that the operator can learn in minutes.

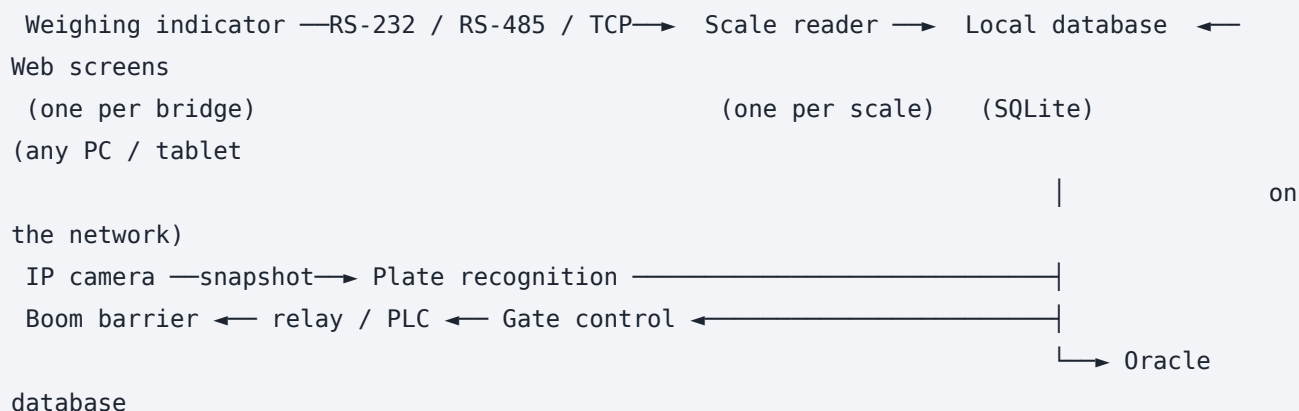
What it does

Area	Capability
Weighing	Reads the weight directly from the indicator (RS-232, RS-485, TCP or Modbus), accepts it only when stable , prints a weighment slip, keeps a permanent record. Two-pass (gross then tare, or tare then gross) or single-pass using a stored vehicle tare.
Several bridges	Any number of scales on one PC, each with its own port, settings and camera. A truck may be weighed in on one bridge and out on another.
Plate recognition	Optional. Reads the vehicle number from a camera, fills the form, and checks that the truck on the bridge is the truck on the ticket.
Gate entry	Register vehicles at the gate, print a gate pass, keep a live list of vehicles inside, block problem vehicles, control the exit.
Gate barriers	OPEN / CLOSE buttons for every boom barrier; barriers can open automatically when a weighing completes.
Oracle	Completed tickets are sent to your Oracle table automatically, safely and without duplicates, even if the network is temporarily down.
Control	Users and roles, audit trail, reports with CSV export, backups, a read-only interface for ERP systems.

Important limits

- The system records what the **indicator** sends. If the weights are used for commercial billing, the indicator and weighbridge must be legally approved and sealed for trade in your country. This software does not replace calibration or statutory certification.
- The software sends only OPEN/CLOSE contacts to a barrier. **Anti-crush safety** (safety loop, photocell, limit switches, emergency stop) must be provided by the barrier hardware. See section 8.

2. System overview



The program runs on **one PC (the "weighbridge server")** next to the weighbridge. Operators use it from a web browser — on that PC, or from any other PC, laptop or tablet on the network. There are two background programs:

- **Scale supervisor** – starts a reader for every enabled scale, restarts it if it stops, and closes gates after their auto-close time.
- **Oracle sync** (*optional*) – sends completed tickets to Oracle every 30 seconds.

Screens

Screen	Who	Purpose
Weighing	Operator	Live weight of each scale, create tickets, second weighment
Gate	Security	Gate OPEN/CLOSE, register vehicles, vehicles inside, exit
Reports	Everyone	Find tickets, totals, CSV, Oracle status
Masters	Everyone (delete/block: admin)	Vehicles (with stored tare), parties, materials
Setup	Administrator	Scales, gates, plate recognition, Oracle, company details, users, diagnostics

Roles

Can do	Operator	Administrator
Weigh, print slips, register vehicles, open/close gates, view reports	✓	✓
Add vehicles, parties, materials	✓	✓
Cancel a ticket or gate entry, delete master data, block vehicles		✓
Override a blocked vehicle, plate mismatch or exit rule (reason is recorded)		✓
Setup page, users, send to Oracle now		✓

3. What you need

3.1 Computer

Item	Requirement
PC	Any modern PC or industrial PC, 4 GB RAM, 10 GB free disk. Windows 10/11, Windows Server 2016+, or Linux (Ubuntu 20.04+ / Debian 11+).
Runs 24×7?	Recommended: yes, with a UPS. The PC must be on for weighing and gate control.
Network	Needed for other PCs/tablets, cameras, network relays and Oracle.

3.2 Software (installed by you or the integrator)

Software	Notes
PHP 8.1 or newer	With extensions <code>pdo_sqlite</code> , <code>sodium</code> , <code>mbstring</code> (required) and <code>curl</code> (needed for cameras, plate recognition and HTTP gate relays).
Web browser	Chrome, Edge or Firefox (current).
Oracle client (<i>optional</i>)	PHP <code>oci8</code> extension and Oracle Instant Client, only for Oracle integration.
Plate recognition service (<i>optional</i>)	CodeProject.AI Server, Plate Recognizer, or OpenALPR.

There is nothing else to install: no separate database server, no Composer, no Node.js.

3.3 Field hardware

Item	Notes
Weighing indicator	Must have a serial output (RS-232 or RS-485) or a network gateway. Note its baud rate, data bits, parity and stop bits (typically 9600, 8, N, 1).
Cable / adapter	RS-232: TX, RX, GND (DB9 pins 2, 3, 5). USB-RS232 or USB-RS485 adapter if the PC has no COM port. RS-485: two wires A and B (D+ and D-) plus a 120 Ω terminator on long runs.
Serial-to-Ethernet gateway (<i>optional</i>)	For distances >15 m (RS-232) or when the PC is far from the bridge.
IP camera (<i>optional</i>)	Must provide a JPEG snapshot URL . One per scale and/or gate.
Boom barrier controller and relay (<i>optional</i>)	A dry-contact relay board or PLC output that can be wired to the barrier's OPEN/CLOSE inputs. IP relays, USB relay boards and Modbus relay modules are supported.

4. Installation

Follow **either** 4.1 (Windows) or 4.2 (Linux). Then continue with section 5.

4.1 Windows

1. **Install PHP.** Download *PHP 8.2 or newer, Thread Safe, x64* (zip) from windows.php.net and unzip to `C:\php`.
2. **Enable extensions.** Copy `C:\php\php.ini-production` to `C:\php\php.ini`. Open it in Notepad and make sure these lines are present without a leading semicolon:
`extension_dir = "ext" extension=pdo_sqlite extension=sodium extension=mbstring extension=curl` Add `C:\php` to the system **PATH** (Windows Settings → search "environment variables").
3. **Copy the program.** Copy the whole `weighbridge` folder to `C:\weighbridge`.
4. **Check the PC.** Open *Command Prompt* and run: `cd C:\weighbridge php bin\check.php` Every line must show `[OK]` (or a `[WARN]` you understand — for example "Oracle driver not installed" if you do not use Oracle). Fix any `[FAIL]` first.
5. **Quick test.** Double-click `start.bat`. Two windows open (readers and web server). Open `http://localhost:8080/` in the browser. Continue with section 5.
6. **Make it permanent (services).** Install *NSSM* (nssm.cc), edit the two paths at the top of `deploy\windows-install.bat`, and run it **as Administrator**. It creates three Windows services that start with the PC: *WeighbridgeScale* (readers), *WeighbridgeOracleSync* (Oracle transfer) and *WeighbridgeWeb* (screens). Close the `start.bat` windows first.
7. **Windows Firewall.** Allow inbound TCP port 8080 if other PCs will connect.

4.2 Linux (Ubuntu / Debian)

```
sudo apt update
sudo apt install -y php-cli php-sqlite3 php-mbstring php-curl nginx php-fpm # or
apache2 + libapache2-mod-php
sudo mkdir -p /var/www && sudo cp -r weighbridge /var/www/weighbridge
sudo chown -R www-data:www-data /var/www/weighbridge/data
sudo chmod 750 /var/www/weighbridge/data
sudo usermod -aG dialout www-data # permission to use serial ports
cd /var/www/weighbridge && sudo -u www-data php bin/check.php
```

Quick test: `./start.sh` (opens the web server on port 8080 and starts the readers).

Permanent:

```
sudo cp deploy/*.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable --now weighbridge-scale weighbridge-oracle-sync
```

and publish `public/` with nginx/Apache. Example nginx site:

```
server {
    listen 80; server_name weighbridge.local;
    root /var/www/weighbridge/public; index index.php;
    location / { try_files $uri $uri/ =404; }
    location ~ /\.php$ { include snippets/fastcgi-php.conf; fastcgi_pass
unix:/run/php/php8.2-fpm.sock; }
    location ~ /\. { deny all; }
}
```

Important: the web server must point at the `public` folder only. The `data` folder (database and key) must never be reachable from the web.

4.3 Folder contents

Path	Contents
<code>public/</code>	The screens (the only folder the web server publishes)
<code>src/</code> , <code>bin/</code>	Program code and background programs
<code>data/</code>	Your data: <code>weighbridge.sqlite</code> (database), <code>app.key</code> (encryption key), <code>snapshots/</code> (camera photos), <code>logs/</code> (reader logs), <code>backups/</code>
<code>sql/oracle_schema.sql</code>	Oracle table script
<code>deploy/</code>	Service files for Linux and Windows
<code>docs/</code>	This manual, screenshots, comparison notes
<code>tests/</code>	Automatic self-tests
<code>VERSION</code> , <code>CHANGELOG.md</code>	Version information

5. First-time setup (step by step)

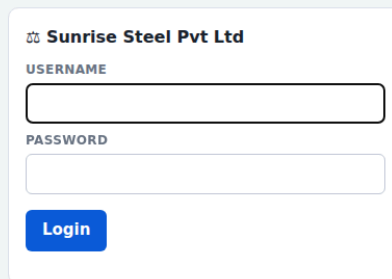
Step 1 — Installer page

Open `http://<server>:8080/install.php`. The page checks the PC (PHP version and extensions, writable data folder). Enter:

- **Company name** (printed on slips)
- **Administrator username and password** (at least 8 characters — keep it safe)

Press **Install**. The system starts with one **Simulator** scale, so you can try everything before connecting real hardware.

Step 2 — Log in



Step 3 — Company details

Setup → General: company name, address (printed on the ticket), ticket prefix (default `WB`, giving numbers such as `WB202609290001`). Tick **Allow manual weight entry** only if you want operators to be able to type a weight when a scale is down (it is flagged on the slip).

Step 4 — Users

Setup → Users: add one **Operator** login per person (never share the administrator login). Minimum password length is 8.

Step 5 — Connect the real scale(s)

See section 6. Switch the scale from *Simulator* to the real port, test, and save.

Step 6 — Masters

Masters: add your regular **vehicles** (with a **stored tare** if you use single-pass weighing), **parties** and **materials**. New names typed on the weighing or gate screens are added automatically.

Step 7 — Optional modules

- Cameras / plate recognition: section 7
- Gates and barriers: section 8
- Oracle: section 9

Step 8 — Acceptance test

Complete the checklist in **Appendix D** before handing over.

6. Connecting the weighing scales (indicators)

Open **Setup** → **Scales**. Every scale on this PC is listed with its connection and reader status.

The screenshot shows the 'Scales' configuration page in the 'Setup' menu. The page has a top navigation bar with 'Sunrise Steel Pvt Ltd' and various menu items. Below the navigation bar, there are tabs for 'Scales', 'Gates', 'Plate recognition', 'Oracle transfer', 'General', 'Users', and 'Diagnostics'. The 'Scales' tab is active, displaying a table titled 'Scales on this PC'. The table has columns for '#', 'NAME', 'CONNECTION', 'CAMERA', and 'READER'. Two scales are listed: 'Main bridge (Scale 1)' and 'North bridge (Scale 2)'. Both are connected via 'Simulator - ascii' and have a camera status of '-'. The reader status for both is '28,450 kg'. Each row has 'Edit', 'Disable', and 'Delete' buttons. Below the table, there is a note: 'Each enabled scale gets its own reader process, started by bin/scale_supervisor.php. Every scale needs its own COM/tty port (or gateway address)'. At the bottom, there is a form titled 'Add a scale' with a text input field for 'NAME (E.G. "GATE 2 BRIDGE")' and an 'Add' button.

#	NAME	CONNECTION	CAMERA	READER	
1	Main bridge (Scale 1)	Simulator - ascii	-	28,450 kg	Edit Disable Delete
2	North bridge (Scale 2)	Simulator - ascii	-	28,450 kg	Edit Disable Delete

Each enabled scale gets its own reader process, started by bin/scale_supervisor.php. Every scale needs its own COM/tty port (or gateway address).

Add a scale

NAME (E.G. "GATE 2 BRIDGE")

Add

- **Add a scale** creates a new one (disabled, in Simulator mode). Press **Edit**, configure it, tick **Enabled**, and **Save**.
- A scale's reader starts within about two seconds of being enabled; changes to a running scale are picked up within five seconds. No restart is needed.
- Two scales cannot use the same COM port or gateway address.

Sunrise Steel Pvt Ltd

Weighing
Gate
Reports
Masters
Setup

admin (admin) · Logout

Scales
Gates
Plate recognition
Oracle transfer
General
Users
Diagnostics

Scale #1

NAME
STATUS

Main bridge (Scale 1)
☒ Enabled

CONNECTION TYPE

Simulator (no hardware, for testing)

PROTOCOL

ASCII continuous output (most indicators)

MINIMUM WEIGHT TO ACCEPT (KG)

20

Camera for this scale (optional)

SNAPSHOT URL (JPEG)

e.g. Hikvision http://192.168.1.64/ISAPI/Streaming/channels/101/picture. Used for the evidence photo and plate recognition.

CAMERA USER
CAMERA PASSWORD

ASCII frame & weight parsing

FRAME TERMINATOR

\r\n

Escapes allowed: \r\n = CR LF, \n, \r, \x03 (ETX).

REQUEST COMMAND (OPTIONAL - POLL-MODE INDICATORS)

Sent before each read, e.g. \r\n. Leave empty for continuous output.

WEIGHT REGEX (GROUP 1 = NUMBER, GROUP 2 = UNIT)

/([+-]?[0-9]+(?:\.[0-9]+)?)([kg|t|lb|oz])?/

Default handles ST,GS,+0012340kg, 12340 kg, =0012340. Use the test below to check.

STABLE REGEX

MOTION REGEX

If your indicator sends a flag, e.g. Stable /\bST\b/, Motion /\bUS\b/. Otherwise leave both empty and the software detects stability itself.

SAME READING N TIMES

TOLERANCE (KG)

5

0

DIVISOR (IMPLIED DECIMALS)

DEFAULT UNIT

1

kg

Save
Back
Test connection (3 s)
Test camera + plate read

Live data from this scale's reader

20,000 kg

MOTION

RUNNING

Raw: US,GS,+0020000kg

6.1 Connection

Field	Meaning
Connection type	<i>Serial</i> (COM / ttyUSB — RS-232 or RS-485 through a USB adapter), <i>TCP/IP</i> (serial-to-Ethernet gateway in "TCP server" mode), or <i>Simulator</i> .
Serial port	Windows COM3 ; Linux /dev/ttyUSB0 (prefer /dev/serial/by-id/... , which does not change after a reboot).
Baud / data bits / parity / stop bits	Must match the indicator exactly (see its manual).
Protocol	<i>ASCII continuous output</i> (most indicators) or <i>Modbus RTU</i> (RS-485 transmitters / load-cell amplifiers).
Minimum weight to accept	Weights below this are refused (prevents empty-platform tickets).

6.2 ASCII protocol

The indicator sends a line of text many times per second, for example `ST,GS,+0012340kg` followed by CR LF.

Field	Meaning
Frame terminator	Character(s) that end each line: <code>\r\n</code> (CR LF, most common), <code>\r</code> , <code>\n</code> , or <code>\x03</code> (ETX).
Request command	Only for indicators that send a weight <i>when asked</i> : the command to send (for example <code>W\r\n</code>). Leave empty for continuous output.
Weight regex	A pattern that finds the number (group 1) and unit (group 2) in the line. The default <code>/([+-]?[0-9]+(?:\.[0-9]+)?)([a-zA-Z]*)(kg g t)?/i</code> handles most formats.
Stable / Motion regex	If the indicator marks stability in the text (for example <code>ST / US</code>), enter e.g. <code>/\bST\b/</code> and <code>/\bUS\b/</code> .
Same reading N times, tolerance	If the indicator has no stability flag, the weight is called stable when the last <i>N</i> readings agree within the tolerance.
Divisor	Use when the indicator omits the decimal point (<code>012340</code> meaning <code>1234.0</code> → divisor 10).
Default unit	Used when the line has no unit.

6.3 Modbus RTU (RS-485)

Slave ID, function (03 holding / 04 input registers), start address (0-based: register 40001 = address 0), one or two registers (16-bit / 32-bit), word order, signed, poll interval. The divisor field (in the ASCII panel) also applies.

6.4 Test

1. **Save**, then press **Test connection (3 s)**. You will see the raw characters received (`<0D>` `<0A>` means CR LF) and the weight the system extracted.
2. If "Could not parse a weight" appears, adjust the terminator or regex until the weight is right.
3. The test needs exclusive use of the port. If the scale is already running, the page tells you: either watch the **Live data** panel at the bottom of the page (it shows what the running reader receives), or disable the scale, wait ten seconds, and test.
4. Put a **known weight** on the bridge and confirm the number on the Weighing screen.

6.5 Wiring notes

- **RS-232:** indicator TX → PC RX, indicator RX → PC TX, GND ↔ GND. If nothing arrives, swap TX/RX.
- **RS-485:** A→A (D+), B→B (D−). Labels differ between vendors; if there is no data, swap the two wires. Add a 120 Ω resistor at the far end of long cables.
- **Indicator setting:** make sure its serial output is set to *continuous* (or the mode you configured), at the same baud/parity.

7. Cameras and plate recognition (ANPR)

Plate recognition is **optional**. Without it, the operator types the vehicle number.

7.1 Cameras

For every scale (*Setup → Scales → Edit → Camera*) and every gate (*Setup → Gates → Edit → Camera*) you can enter a **snapshot URL** — a web address that returns the current picture as a JPEG — with the camera's user name and password. Typical examples: Hikvision `http://<ip>/ISAPI/Streaming/channels/101/picture`, Dahua `http://<ip>/cgi-bin/snapshot.cgi`. Check your camera manual.

With a camera configured, an **evidence photo** is saved at every first and second weighment and at every gate entry/exit, and shown on the slip / gate pass.

Camera placement (for plate reading): 5–8 m from the plate, angle less than 30°, shutter at least 1/500 s, good lighting or infra-red at night, no headlight glare.

7.2 Recognition service

Setup → Plate recognition. Choose a provider:

Provider	Notes
CodeProject.AI Server	Free, runs on your own PC or server. Install it and its ALPR module; URL <code>http://127.0.0.1:32168/v1/vision/alpr</code> .
Plate Recognizer	Commercial. Cloud service (needs internet and an API token) or their on-premises container.
Local command	Any program that prints the plate as JSON or as <code>PLATE / PLATE 0.93</code> , for example OpenALPR: <code>alpr -c in -n 1 -j {image}</code> . Run without a shell for safety.

The screenshot shows the 'Setup' page for 'Sunrise Steel Pvt Ltd'. The 'Plate recognition' tab is selected. The configuration section includes:

- Automatic number-plate recognition (ANPR)**: A note states 'A camera per scale (Setup > Scales) supplies the picture. The recognition itself runs in a service you choose:'.
- PROVIDER**: A dropdown menu currently set to 'Off'.
- MINIMUM CONFIDENCE (0-1)**: A text input field containing '0.6'.
- WHEN PLATE DIFFERS FROM TICKET VEHICLE**: A dropdown menu set to 'Flag ticket (MISMATCH) + audit'.
- ☐ Auto-read the plate when a truck settles on the selected scale and the vehicle box is empty.
- Save**: A blue button.
- Test**: A section header.
- UPLOAD A SAMPLE PICTURE OF A PLATE (JPEG)**: A file upload area with a 'Choose File' button and 'No file chosen' text.
- Recognise sample (uses the values above, even if unsaved)**: A button.
- A note at the bottom: 'Accuracy depends on camera position, lighting and the ANPR service. Test with your own gate pictures before relying on *Block*.'

- **Minimum confidence** (default 0.6): a plate read below this is treated as *unread* rather than risk a wrong number.
- **When the plate differs from the ticket vehicle**: *Ignore*, *Flag ticket* (recorded as MISMATCH, audit entry), or *Block* (an administrator must tick **Override**).
- **Auto-read**: when a truck settles on the selected scale and the vehicle box is empty, the plate is read automatically.
- **Test**: upload a sample photo of a plate and press *Recognise sample* — it works with unsaved values. On a scale's page, **Test camera + plate read** checks the real camera.

7.3 How it is used

- **Read** button next to the vehicle number: fills the number, tells you whether the vehicle is known, its stored tare, and whether it already has an open ticket (or is on the block list, at the gate).
- On the **second weighment** the system checks that the truck on the bridge is the truck on the ticket.
- Results are stored on the ticket/pass and shown as **OK**, **MISMATCH**, **UNREAD** (plate not readable) or **NOIMG** (no camera picture).
- Plate comparison ignores spaces and dashes and look-alike characters (0/O, 1/I, 5/S, 8/B, 2/Z), and tolerates one wrong character on plates of six or more characters.
- A camera or recognition failure **never stops a weighing** — the ticket is created and flagged.

Advice: test on your own pictures first and start with *Flag*, not *Block*.

8. Gates and boom barriers

8.1 Safety first

The program only sends a momentary **contact** (like a remote-control button press) to the barrier controller. It cannot see the barrier and does **not** provide anti-crush protection.

- Fit the barrier's own **safety loop / photocell / limit switches / emergency stop** as required.
- Connect through a **dry-contact relay** to the controller's OPEN/CLOSE inputs. Never connect PC outputs directly to a motor circuit.
- **Auto-close** is **off** by default. Use it only when a safety loop is fitted.
- Test with people and vehicles clear of the barrier.
- The status badge shows the **last command sent**, not the real barrier position.

8.2 Configure a gate

Setup → Gates → Add a gate → Edit.

Sunrise Steel Pvt Ltd

WeighingGateReportsMastersSetup

admin (admin) · Logout

ScalesGatesPlate recognitionOracle transferGeneralUsersDiagnostics

Gate #1

NAME

Main gate IN

STATUS

☒ Enabled

ROLE

Entry gate

LINKED SCALE (WEIGHBRIDGE BOOMS)

Any scale

Howbridge exit boom: opens automatically when a weighing on the linked scale completes.

HOW IS THE BARRIER CONTROLLED?

Simulator (no hardware)

ACTION TYPE

Pulse (momentary contact, th

PULSE LENGTH (MS)

1000

AUTO-CLOSE AFTER (S, 0 = OFF)

0

AUTOMATION

☒ Allow automatic opening (weighing done)

Auto-close runs in the supervisor service. Only enable it with a safety loop fitted.

Camera at this gate (optional, for plate reading)

SNAPSHOT URL (JPEG)

CAMERA USER

CAMERA PASSWORD

Connection

Simulator: commands are only recorded. Use it to try the screens.

Save

Back

Test OPEN (uses the values above)

Test CLOSE

Field	Meaning
Role	<i>Entry gate / Exit gate (used at the security gate), Weighbridge entry boom, Weighbridge exit boom.</i>
Linked scale	For a weighbridge exit boom: which scale it belongs to (or "Any").
Action type	Pulse (momentary contact, then released — most barrier controllers) or Latched (relay stays on until the next command).
Pulse length	Milliseconds the contact is held (typical 500-1000).
Auto-close after	Seconds after OPEN before a CLOSE is sent (0 = never). Performed by the scale supervisor service, which must be running.
Allow automatic opening	Lets the system open this gate by itself (weighbridge exit boom opens when a weighing completes).

How the barrier is controlled

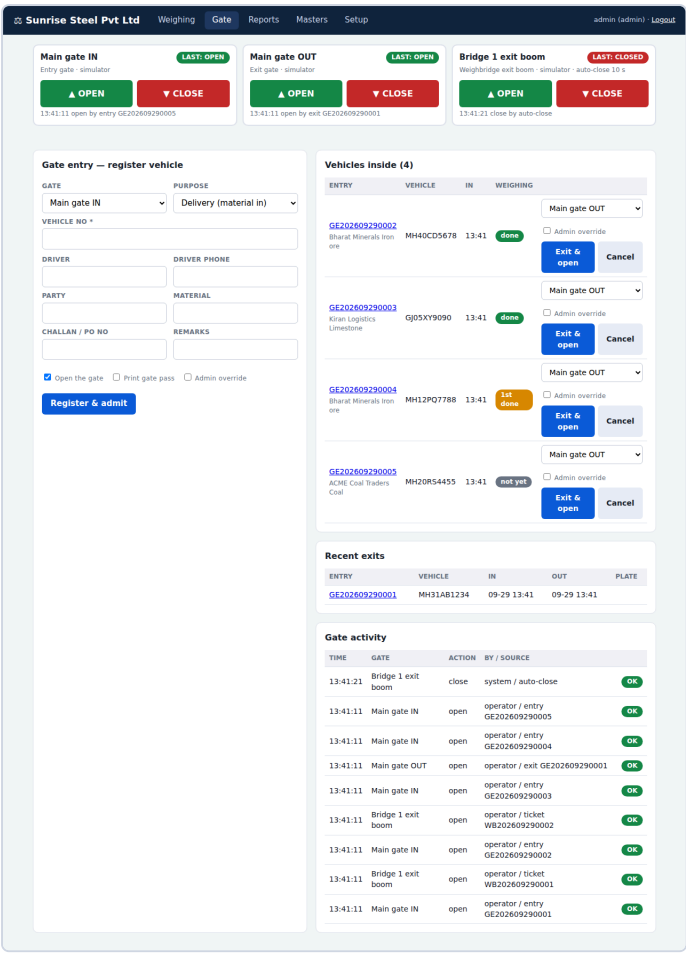
Hardware	Driver	Typical settings
IP relay, Shelly, ESP relay, barrier with web API	HTTP	Open URL, Close URL, Release URL (optional), method GET/POST/PUT, optional user/password and body. Success = HTTP 2xx. Example Shelly: open <code>http://<ip>/relay/0?turn=on</code> , release <code>http://<ip>/relay/0?turn=off</code> .
Ethernet relay board	TCP	Host, port, open / close / release commands as text (<code>OPEN\r\n</code>) or hex (<code>hex:A0 01 01 A2</code>).
USB / RS-232 relay board	Serial	Its own COM port, baud (usually 9600); commands as hex, e.g. open <code>hex:A0 01 01 A2</code> , release <code>hex:A0 01 00 A1</code> .
RS-485 relay module / PLC	Modbus RTU	Slave ID, open coil, optional close coil. Needs its own adapter.
PLC / Ethernet I/O module	Modbus TCP	Host, port 502, unit ID, coils.
—	Simulator	Commands are only recorded; for training.

Modbus coils: *one coil, latched* = ON opens, OFF closes. *Two coils, pulse* = each coil is pulsed.

A gate relay cannot share a COM port with a scale. Two gates may share one relay board (different channels).

Use **Test OPEN** / **Test CLOSE** in the editor (it works before you save) and watch the barrier.

8.3 The Gate screen



- 1. **Gate cards** — big **OPEN** and **CLOSE** buttons. Every press is logged (who, when, result). If the relay does not answer, the card shows the failure and the state does not change.
- 2. **Gate entry — register vehicle** (section 11).
- 3. **Vehicles inside** and **Recent exits**.
- 4. **Gate activity** — the last commands, including automatic ones (auto-close , ticket WB... , entry GE...).

8.4 Automatic actions

Event	Automatic action
Vehicle registered at an entry gate (with "Open the gate" ticked)	Entry gate opens
Vehicle exits (with "Exit & open")	Exit gate opens
Second weighment (or single-pass ticket) completes on a scale	Weighbridge exit boom linked to that scale opens (if "Allow automatic opening" is ticked)
Auto-close time passes	Gate closes

A barrier fault **never** blocks or undoes a weighing — the failure is shown in Gate activity.

8.5 Gate rules

Setup → Gates → Gate entry rules:

- **Weighing requires a gate entry** — *No / Warn / Yes - block* (an administrator can override).
- **Exit requires completed weighment and no open ticket** — *No / Warn / Yes - block*.

9. Oracle database integration

9.1 What is sent

Every **completed** (closed) ticket, and every **cancelled** ticket that was already sent, is transferred to one Oracle table (default `WEIGHBRIDGE_TICKETS`). The transfer uses an Oracle `MERGE` on the ticket number, so a ticket that is sent twice updates the same row — **no duplicates**. If Oracle or the network is down, tickets simply wait (status `PENDING`) and are sent automatically when it returns. Weighing is never interrupted.

9.2 Prepare Oracle

1. Run `sql/oracle_schema.sql` in the schema that will own the table (creates `WEIGHBRIDGE_TICKETS`, primary key `TICKET_NO`, two indexes).
2. Create a dedicated user with only `CREATE SESSION` and `SELECT, INSERT, UPDATE` on that table.
3. If you created the table with an earlier script, run the `ALTER TABLE` statement at the bottom of the file to add `SCALE_NAME`, `PLATE_IN`, `PLATE_OUT`, `PLATE_FLAG`.
4. Make sure the weighbridge PC can reach the listener (default port 1521).

9.3 Prepare the PC

Install **Oracle Instant Client (Basic)** matching PHP's 32/64 bit, and the PHP **oci8** extension (README step 6 has the exact Windows and Linux commands). Check with `php -m` that `oci8` is listed, and run `php bin/check.php`.

9.4 Configure

Setup → Oracle transfer: host, port, **service name** (not SID), user, password (stored encrypted), table name, batch size. Press **Test Oracle connection** (it also checks that the table exists), tick **Enable automatic transfer**, and **Save**. The *WeighbridgeOracleSync* service (or `bin/sync_oracle.php --loop=30`) performs the transfer.

Weighing

Gate

Reports

Masters

Setup

admin (admin) · [Logout](#)

Scales

Gates

Plate recognition

Oracle transfer

General

Users

Diagnostics

Oracle database

PHP Oracle driver:

NOT INSTALLED

Install oci8 + Instant Client (README step 6) before enabling transfer.

☐ Enable automatic transfer of completed tickets to Oracle

HOST

PORT

1521

SERVICE NAME (NOT SID)

USER

PASSWORD

TARGET TABLE

BATCH SIZE

WEIGHBRIDGE_TICKETS

50

Create the table with `sql/oracle_schema.sql`. Tickets are pushed with MERGE keyed on TICKET_NO, so retries never duplicate. If Oracle is down tickets stay PENDING and are sent when it returns.

Save

Test Oracle connection

Send pending now

9.5 Monitoring

- *Reports* shows the Oracle status of each ticket: **PENDING** (waiting), **SYNCED**, **FAILED** (hover for the Oracle error), **NA** (Oracle not enabled).
- Administrators can press **Send to Oracle now** or **Re-queue failed**.
- *Setup* → *Diagnostics* shows the number waiting and the last error.
- `http://<server>/api/health.php` returns `503` when a scale reader is not running — use it with your monitoring system.

9.6 Columns

See Appendix B.

10. Daily operation — weighbridge operator

Sunrise Steel Pvt Ltd

Weighing

Gate

Reports

Masters

Setup

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Main bridge (Scale 1)

STABLE

28,450 kg

RUNNING

North bridge (Scale 2)

STABLE

28,450 kg

RUNNING

New ticket — first weighment

VEHICLE NO *

DIRECTION

Inward (receive)

PARTY

MATERIAL

DRIVER

CHALLAN / PO NO

THIS WEIGHMENT IS

GROSS (loaded) - tare later

OPTIONS

☐ Use stored tare (single pass, gross now)

REMARKS

Capture weight & create ticket

Open tickets (waiting for second weighment)

TICKET	VEHICLE	1ST KG	
WB202609290004	MH12PQ7788	30,110 GROSS	2nd weight Cancel

Recent tickets

TICKET	VEHICLE	NET KG	PLATE	ORACLE
WB202609290003	GJ05XY9090	18,410		NA
WB202609290002	MH40CD5678	20,120		NA
WB202609290001	MH31AB1234	22,100		NA

10.1 The screen

- At the top, one **live display per scale**: weight in kg and a badge **STABLE** (green), **MOTION** (amber) or **NO DATA** (red). Click the display of the bridge you are standing at; the choice is remembered on that PC.
- The **capture buttons are disabled until the weight is stable**. The weight recorded is the one the system holds at the moment you press the button — it cannot be typed or altered in the browser.
- Left: **New ticket**. Right: **Open tickets** (waiting for the second weighment) and **Recent tickets**.

10.2 Normal two-pass weighing (inward)

- Truck drives onto the bridge. Wait for **STABLE**.
- Enter the **vehicle number** (or press **Read** to read the plate). Choose party and material; driver and challan are optional.
- Leave *This weighment is* = **GROSS (loaded) - tare later** for a loaded truck arriving, or choose **TARE (empty)** for an empty truck arriving to load.
- Press **Capture weight & create ticket**. The ticket appears under *Open tickets*.
- Truck goes away, unloads or loads, and returns.
- On the returning truck's row press **2nd weight** (with the truck stable on a scale — it can be a different bridge; select it at the top).

- The slip opens. The system computes **Net = Gross – Tare**, shows all values, and prints on **Print**. The weighbridge exit boom opens if configured.

The system refuses a second weightment if Gross is not greater than Tare (wrong first-weight type) — cancel the ticket (administrator) and redo.

10.3 Single-pass weighing (stored tare)

For vehicles whose empty weight is stored in *Masters*: tick **Use stored tare (single pass, gross now)**, capture the loaded weight; the ticket is completed immediately. The stored tare is learned automatically from the first two-pass weighing of a vehicle.

10.4 Slip

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Sunrise Steel Pvt Ltd

Plot 14, MIDC Industrial Area, Nagpur 441108

WEIGHTMENT SLIP

TICKET NO	WB202609290001	DIRECTION	INWARD
VEHICLE	MH31AB1234	DRIVER	
PARTY	ACME Coal Traders	MATERIAL	Coal
SCALE IN / OUT	Main bridge (Scale 1) / Main bridge (Scale 1)	PLATE CHECK	-
CHALLAN	CH-7006	OPERATOR	operator
1ST (GROSS)	31,250 kg 2026-09-29 13:41:11	2ND	9,150 kg 2026-09-29 13:41:11
GROSS	31,250 kg	TARE	9,150 kg
NET WEIGHT	22,100 kg		

Print
Back

10.5 Messages you may see

Message	Meaning / action
"Weight is not stable yet"	Wait for the STABLE badge.
"No live weight ... reader stopped or scale disconnected"	Check the indicator and cable; ask the administrator to look at <i>Setup</i> → <i>Scales</i> .
"Weight ... is below the minimum capture weight"	Platform empty or truck partly off the bridge.
"Vehicle ... already has an open ticket"	Use 2nd weight on the open ticket instead of creating a new one.

Message	Meaning / action
"Gross must be greater than tare"	The first-weight type was wrong; administrator cancels and redoes.
"Camera read plate X but the ticket vehicle is Y"	Plate check (block mode): confirm the vehicle number; only an administrator can override.
"Vehicle ... has no gate entry"	Gate rule: register the vehicle at the gate first (or administrator override).

10.6 If the scale is down

If the administrator has allowed **manual weight entry**, a *Manual weight* box appears; enter the weight from another source. Such weights are marked (*manual*) on the slip and should be exceptional.

10.7 Cancelling

Only an administrator can **Cancel** an open ticket (a reason is required and is recorded).

11. Daily operation — security gate

Open the **Gate** screen.

11.1 Admitting a vehicle

1. Choose the **entry gate** and **purpose** (delivery, dispatch, visitor/other, empty/returning).
2. Enter the **vehicle number**, or press **Read** to read it from the gate camera. The note under the field tells you if the vehicle is *known*, *new*, **blocked**, or **already inside**.
3. Fill in driver, phone, party, material, challan.
4. Leave **Open the gate** ticked; tick **Print gate pass** if you want a printout.
5. Press **Register & admit**.

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Sunrise Steel Pvt Ltd

GATE PASS

ENTRY NO	GE202609290001	STATUS	EXITED
VEHICLE	MH31AB1234	PURPOSE	Delivery (material in)
DRIVER	Suresh Patil 98220 12345	CHALLAN	CH-1402
PARTY	ACME Coal Traders	MATERIAL	Coal
IN	2026-09-29 13:41:11 Main gate IN · operator	OUT	2026-09-29 13:41:11 Main gate OUT · operator
WEIGHMENT	WB202609290001 (CLOSED, net 22,100 kg)		

Print

Back

Refusals: a vehicle that is already inside, or on the **block list**, is refused and the gate stays closed (the attempt is recorded). An administrator can admit a blocked vehicle by ticking *Admin override*; this is noted on the pass.

11.2 Vehicles inside

The list shows every vehicle currently inside with its **weighing status**: *not yet*, *1st done* (loaded/empty weight taken, second pending) or *done*.

11.3 Letting a vehicle out

1. Find the vehicle under *Vehicles inside*.
2. Check the exit gate in the drop-down and press **Exit & open**.
3. If the exit rule is set to *block*, the exit is refused while the vehicle still has an **open ticket** or (for delivery/dispatch) **no completed weighment**. The message says which. An administrator may override; it is recorded on the pass.

11.4 Opening or closing a barrier manually

Press **OPEN** or **CLOSE** on the gate's card. The result appears under the buttons and in *Gate activity*. Use manual buttons for staff, emergency vehicles and testing.

11.5 Blocking a vehicle

Masters → *Vehicles* → *Block* (administrator): enter a reason. The vehicle is refused at the gate until unblocked.

12. Reports, tickets and the ERP interface

12.1 Reports

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FROM01/01/2000TO01/01/2100SEARCHvehicle / party / material /ORACLEAllFilterCSV

0 ticket(s) waiting for OracleSend to Oracle nowRe-queue failed

4 tickets · Net total 60,630 kg (60.630 t)

TICKET	DATE	VEHICLE	PARTY	MATERIAL	SCALE	GROSS	TARE	NET	STATUS	PLATE	ORACLE
WB202609290004	2026-09-29 13:41	MH12PQ7788	Bharat Minerals	Iron ore	Main bridge (Scale 1)				OPEN		NA
WB202609290003	2026-09-29 13:41	GJ05XY9090	Kiran Logistics	Limestone	North bridge (Scale 2)	27,400	8,990	18,410	CLOSED		NA
WB202609290002	2026-09-29 13:41	MH40CD5678	Bharat Minerals	Iron ore	Main bridge (Scale 1)	28,940	8,820	20,120	CLOSED		NA
WB202609290001	2026-09-29 13:41	MH31AB1234	ACME Coal Traders	Coal	Main bridge (Scale 1)	31,250	9,150	22,100	CLOSED		NA

Reports lists tickets for a date range with search (vehicle, party, material, ticket number) and an **Oracle** filter. It shows totals in kg and tonnes for completed tickets, plate check result, scale, and Oracle status. **CSV** downloads the filtered list for Excel. Click a ticket number to view or reprint the slip.

12.2 Masters

Sunrise Steel Pvt Ltd

WeighingGateReportsMastersSetup

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VehiclesPartiesMaterials

Add Vehicles

REGISTRATION NO

STORED TARE KG (OPTIONAL)

TRANSPORTER

Add

VEHICLE	TARE KG	TRANSPORTER	GATE		
GJ05XY9090	8,990			Block	Delete
MH12PQ7788	-			Block	Delete
MH20RS4455	-			Block	Delete
MH31AB1234	9,150	Kiran Logistics		Block	Delete
MH40CD5678	8,820	Bharat Minerals		Block	Delete
MH49ZZ0001	-		BLOCKED	Unblock	Delete

Vehicles (registration, **stored tare**, transporter, blocked flag), parties and materials. Add anyone; administrators can delete and block.

12.3 ERP interface (read-only REST API)

Setup → General → ERP REST API → Generate new token (the token is shown **once**; store it safely; it is kept only as a hash). Your ERP then calls:

```
GET http://<server>/api/v1/tickets.php?since_id=0&limit=100
Authorization: Bearer <token>
```

It returns completed and cancelled tickets with `id` greater than `since_id` (maximum 500 per call) as JSON. The ERP remembers the highest `id` received and asks for newer ones next time. *Disable API* revokes access.

13. Administration

13.1 Users

Setup → Users: add users, enable/disable, set passwords. The administrator cannot disable their own login. After 8 failed logins for a user name or address, further attempts are refused for 10 minutes.

13.2 Backups — do this daily

```
php bin/backup.php --dir=D:\Backups\weighbridge --keep=30
```

Creates a consistent copy of the database (safe while the system is running), verifies it, and keeps the newest 30. Schedule it daily: Windows *Task Scheduler*, or Linux cron `15 2 * * * php /var/www/weighbridge/bin/backup.php --dir=/backup/weighbridge --keep=30`.

Also keep a separate copy of `data/app.key` — without it the saved Oracle, camera and relay passwords cannot be decrypted (you would re-enter them). Copy the `data/snapshots` folder if you need to keep evidence photos.

13.3 Restoring / moving to another PC

1. Install the program on the new PC (section 4) but do not run the installer page.
2. Stop the services. Copy your backup file to `data/weighbridge.sqlite` (delete any `weighbridge.sqlite-wal/-shm` files) and copy `app.key` to `data/`.
3. Run `php bin/check.php`, start the services, log in.
4. Re-check the COM port names in *Setup → Scales / Gates* (they can differ on a new PC).

13.4 Upgrading to a new version

1. Take a backup (13.2).
2. Stop the services.
3. Copy the new program files over the old ones, **keeping the `data` folder**.
4. Run `php bin/check.php`, then start the services. The database upgrades itself automatically the first time it is opened.
5. Optional: `php tests/run.php` runs the self-tests in a temporary folder (it never touches your data).

13.5 Logs and health

What	Where
Reader logs (one per scale)	<code>data/logs/scale_<n>.log</code> (rotated at 5 MB)
Supervisor / service messages	Linux: <code>journalctl -u weighbridge-scale</code> ; Windows: <code>data\scale.log</code>

What	Where
Audit trail (logins, tickets, cancels, overrides, setup changes)	<i>Setup → Diagnostics → Recent audit</i>
Gate commands	<i>Gate → Gate activity</i>
System status	<i>Setup → Diagnostics, and <code>api/health.php</code></i>

Camera photos in `data/snapshots` are never deleted automatically; remove old ones according to your retention policy.

13.6 Time and date

Tickets use the PC clock and timezone. The default timezone is `Asia/Kolkata`; change it by setting the environment variable `WB_TZ` (for example `WB_TZ=Africa/Nairobi`) for the services.

14. Security

- Passwords are stored as one-way hashes. Oracle, camera and relay passwords and the plate-recognition token are stored **encrypted** (key in `data/app.key`).
- Every form is protected against cross-site request forgery; every database query is parameterised; browser security headers are set; repeated failed logins are locked out.
- Weights are captured **on the server** from the reader; the browser cannot supply a weight (except the optional, flagged manual entry).
- Setup, user management, overrides and cancellations are administrator-only and audited.
- Serial port names and network addresses entered in Setup are validated so they cannot be used to run commands.
- **Recommendations:** keep the PC on a protected network; use HTTPS (reverse proxy) if the screens are reachable beyond the plant network; use individual logins; back up daily; restrict who can open the `data` folder; fit the barrier's hardware safety devices.

15. Troubleshooting

First run `php bin/check.php` and look at *Setup → Diagnostics*.

Symptom	Likely cause and fix
Display shows --- , badge NO DATA	Reader not running (start the service / <code>start.bat</code>), wrong port or settings, cable. Use <i>Setup → Scales → Test connection</i> .
"Serial device not found" / "stty failed"	Wrong port name; on Linux add the service user to <code>dialout</code> ; unplug and re-plug the adapter.
Test: port opens but no data	TX/RX swapped (RS-232), A/B swapped (RS-485), wrong baud/parity, indicator not in continuous mode (use <i>Request command</i>).
Bytes arrive but "could not parse a weight"	Fix terminator (<code>\r</code> , <code>\n</code> , <code>\r\n</code> , <code>\x03</code>) and weight regex.
Weight never becomes STABLE	Enter the indicator's stable flag in the Stable regex, or raise tolerance / lower "same reading N times".
Value 10× too large or small	Set Divisor (10, 100...).
Modbus: "short/no response"	Slave ID, baud/parity, A/B wiring, address base (0-based).
Modbus: CRC error	Baud/parity mismatch, noise, missing terminator resistor.
Windows: port busy	Another program (the indicator's own utility) holds the COM port.
Scale shows "Reader not running"	Supervisor service stopped, or the scale is not Enabled.
"already uses COM3" when saving	Each scale/gate relay needs its own serial port.
Camera test: "No JPEG from the camera URL"	Wrong snapshot URL/credentials, camera unreachable, <code>curl</code> extension disabled.
Plate always UNREAD / NOIMG	Camera or service problem (use the test buttons), poor picture, confidence threshold too high.
Many MISMATCH flags	Camera angle/lighting; set policy to <i>Flag</i> ; try another provider.
Gate button: "relay unreachable" / "did not reply"	Wrong IP/port/COM, wiring, slave ID; use <i>Test OPEN</i> in <i>Setup → Gates</i> . For Modbus boards that never answer, untick <i>Require echo reply</i> .
Log says OK but the barrier does nothing	Wrong relay channel/coil, or the controller needs a longer pulse.
Gate did not auto-close	Supervisor not running, or auto-close is 0.
"No Oracle PHP driver"	Install oci8 + Instant Client (section 9.3); check <code>php -m</code> for the same PHP the services use.
ORA-12541 / ORA-12514	Listener host/port or service name wrong.
ORA-00942	Table missing or no rights: run <code>sql/oracle_schema.sql</code> , grant rights.

Symptom	Likely cause and fix
Tickets stay PENDING	Oracle sync service not running or Oracle unreachable — see <i>Diagnostics</i> .
"Too many failed logins"	Wait 10 minutes, or ask another administrator.
Screens load but look unstyled / links 404	Web server is not pointing at the <code>public</code> folder.

When you ask for support, please send: the output of `php bin/check.php`, a screenshot of *Setup* → *Diagnostics*, the last lines of `data/logs/scale_<n>.log`, the exact message on screen, and the indicator model with its serial output settings.

16. Appendices

Appendix A — Settings reference

Setting	Where	Default
Company name / address / ticket prefix	General	My Weighbridge / — / WB
Allow manual weight	General	off
Connection type, port, baud, data bits, parity, stop bits	Scales	Simulator; 9600, 8, N, 1
Protocol, terminator	Scales	ASCII; \r\n
Weight regex	Scales	/([+-]?[0-9]+(?:\.[0-9]+)?)\s*(kg g t)?/i
Stable regex / motion regex	Scales	empty (use repeated readings)
Same reading N times / tolerance	Scales	5 / 0 kg
Divisor	Scales	1
Minimum weight	Scales	20 kg
Camera URL, user, password	Scales, Gates	empty
Gate driver / mode / pulse / auto-close / automatic opening	Gates	Simulator / pulse / 1000 ms / 0 / on
ANPR provider, URL, token, region	Plate recognition	off
ANPR minimum confidence / policy / auto-read	Plate recognition	0.6 / flag / off
Weighing requires gate entry / exit rule	Gates	off / warn
Oracle host, port, service, user, password, table, batch	Oracle transfer	— / 1521 / — / — / — / WEIGHBRIDGE_TICKETS / 50

Appendix B — Oracle table `WEIGHBRIDGE_TICKETS`

Column	Type	Content
TICKET_NO	VARCHAR2(30), primary key	Ticket number, e.g. WB202609290001
VEHICLE_NO	VARCHAR2(30)	Vehicle registration
PARTY, MATERIAL, DIRECTION, DRIVER, CHALLAN_NO, REMARKS	VARCHAR2	As entered (DIRECTION = INWARD / OUTWARD)
GROSS_KG, TARE_KG, NET_KG	NUMBER(12,2)	Weights in kg

Column	Type	Content
FIRST_WEIGHT_AT , SECOND_WEIGHT_AT	TIMESTAMP	Local time of each weighment
STATUS	VARCHAR2(12)	CLOSED or CANCELLED
OPERATOR	VARCHAR2(50)	User who created the ticket
LOCAL_ID	NUMBER(12)	Record number in the weighbridge database
SCALE_NAME	VARCHAR2(100)	Scale used for the first weighment
PLATE_IN , PLATE_OUT , PLATE_FLAG	VARCHAR2	Plates read, and OK / MISMATCH / UNREAD / NOIMG
SYNCED_AT	TIMESTAMP	When the row was written

Appendix C — Indicator output examples

These are only **examples of the kind of line** indicators send. Always check your indicator's manual and use *Test connection* to see what yours really sends.

Line received (each ends with CR LF)	Works with the default regex?	Note
ST,GS,+0012340kg	✓	Enter Stable regex /\bST\b/ and Motion regex /\bUS\b/ if the line begins with ST/US
12340 kg	✓	
=0012340	✓	Unit defaults to kg
012340 meaning 1234.0 kg	✓ number, ✗ scale	Set Divisor to 10
W 1.5 t	✓	Converted to 1500 kg
Frame <STX> 5000 kg <ETX>	✓	Terminator \x03

Appendix D — Commissioning and acceptance checklist

Site		Date	
Engineer		Client witness	

#	Test	Result
1	php bin/check.php shows no [FAIL]	☐
2	Installer completed; administrator and operator logins work	☐
3	Services start automatically after a PC restart (readers, web, Oracle sync)	☐

#	Test	Result
4	Each scale displays the correct weight; empty bridge reads 0 (or below minimum)	☐
5	Known test weight placed on each bridge reads correctly	☐
6	STABLE appears only when the weight is really steady; buttons are disabled during MOTION	☐
7	Two-pass ticket: gross, tare and net are correct; slip prints	☐
8	Single-pass ticket with stored tare is correct	☐
9	Second weighment on a different bridge works	☐
10	Unplugging the indicator shows NO DATA within 3 s and blocks capture	☐
11	Camera test returns a picture; photo appears on the slip	☐
12	Plate recognition reads at least 8 of 10 real vehicles (or is left off)	☐
13	Each gate: Test OPEN and Test CLOSE move the barrier (people clear); barrier's safety devices verified	☐
14	Gate entry → weighing → exit works; exit refused with an open ticket (if rule is on)	☐
15	Blocked vehicle is refused at the gate	☐
16	Oracle test succeeds; a ticket appears in Oracle with correct values and status SYNCED	☐
17	Oracle down for a while: weighing continues, tickets are sent when it returns	☐
18	Report and CSV export match the slips	☐
19	Backup taken and a test restore checked; <code>app.key</code> copied elsewhere	☐
20	Operators trained; administrator password handed over securely	☐

Appendix E — Main database tables (SQLite)

`weighments` (tickets), `gate_entries` (gate passes), `vehicles`, `parties`, `materials`, `scales`, `live_scale` (current readings), `gates`, `gate_state`, `gate_events`, `users`, `settings`, `audit`, `login_attempts`. The file is `data/weighbridge.sqlite`; it can be opened read-only with any SQLite tool for your own reports. Please do not edit it while the system is running.

Appendix F — Glossary

Term	Meaning
Gross / Tare / Net	Loaded weight / empty weight / material weight (Gross – Tare).
Indicator	The display unit connected to the load cells; it sends the weight to the PC.

Term	Meaning
RS-232 / RS-485	Serial connections. RS-485 uses two wires and works over long distances and multiple devices.
Modbus	A common industrial protocol used by transmitters, relays and PLCs.
Stable	Weight is steady (no motion), safe to record.
ANPR	Automatic number-plate recognition.
Boom barrier	The gate arm at the entry, exit or weighbridge.
Pulse	A short contact closure, like pressing a remote-control button.
Reader / supervisor	Background programs that read a scale / that keep all readers running.
Store-and-forward	Records are kept locally and sent when the destination is available.

End of manual — Weighbridge Management System v1.0.0